Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	*(a)*		Diagram showing relevant right-angled triangle with d and θ labelled (1) $d \sin\theta$ stated to be path difference [between light from adjacent slits] (1) $d \sin\theta = n\lambda$ is condition for constructive interference [or for light to 'arrive' in phase (1) mention of constructive interference only don't award the mark	3			3	1	
	(b)	(i)	Scales correctly chosen (more than ½ used) and orientated and axes clearly labelled (1) $\sin\theta$ plotted correctly, whether or not tabulated - allow ± ½ small square tolerance (1) Best straight line through origin or by eye (1) **If θ plotted:** don't award the scales mark, plotting of points correctly mark can be awarded and best fit curve or straight line can be awarded		3		3	3	3
		(ii)	Gradient = $\frac{\lambda}{d}$ **or** single point on graph taken by implication (1) Gradient = 0.285 [± 0.005] (1) λ = 510 [± 15] nm (1) **unit mark** **If θ plotted:** If λ determined correctly award a maximum of 2 marks			3	3	3	3
			Question 4 total	**3**	**3**	**3**	**9**	**7**	**6**